ETH zürich

Mobile Computing

Emerging Technologies
IEEE 802.15 Bluetooth and Zigbee

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Course Outline

Introduction

1. Data and Signals
2. Data Communication
3. Mobile Systems
4. Mobile Algorithms
5. Mobile AI
6. Cloud Computing

Tournament

Short-Range Connectivity: Who Defines What?

Standards body	Radio / MAC	Higher-layer functionality
Bluetooth SIG	Bluetooth Classic / LE	Bluetooth profiles and services
IEEE + CSA	IEEE 802.15.4	Zigbee networking and applications
IEEE	802.15.4 UWB 802.15.6 body area 802.15.7 optical	Upper layers depend on the technology and application

IEEE 802.15.4 also supports Thread; Matter runs above IP over Thread, Wi-Fi or Ethernet.

IEEE 802.15: Bluetooth Origins & High-Rate

- 802.15.1 - historical Bluetooth standard
  - Modern Bluetooth Classic and LE: Bluetooth SIG
- 802.15.2 - coexistence mechanisms
- 802.15.3 - high-rate WPAN family
  - .3a: competing UWB proposals; withdrawn in 2006
  - .3b: MAC enhancements; .3c: millimeter-wave PHY

IEEE 802.15: Low-Rate and Specialty Networks

- 802.15.4 - PHY/MAC for low-rate networks
  - Zigbee adds network and application layers
  - .4a/.4z: UWB; .4e: industrial MAC; .4g: smart utilities
- 802.15.5 - mesh; 802.15.6 - body area networks
- 802.15.7 - optical wireless communication
  - Other projects: peer awareness, key management, routing

Bluetooth Logo

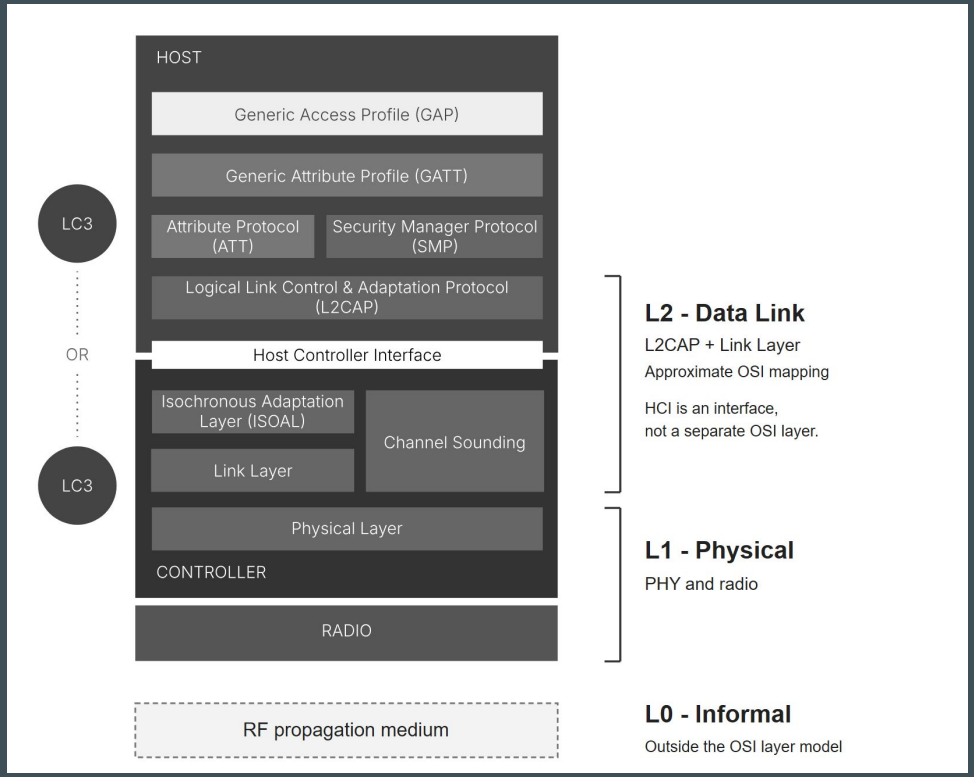
- In 1997, the specification was named after a Danish king
- It should unite devices as King Harald Bluetooth united the tribes of Denmark
- The logo combines Scandinavian runes for the king's initials, H and B



802.15.1 Bluetooth Protocol Stack

- Link Controller (LC)
  - Connection establishment, addressing
  - Packet formatting, timing, and power control
- Link Manager Protocol (LMP)
  - Link setup and link management
  - Encryption, authentication
- Logical Link Control and Adaptation Protocol (L2CAP)
  - Adapts upper layer protocols to baseband layer
  - Provides connectionless and connection oriented services

Bluetooth LE Protocol Stack and OSI Reference



L7 Application

L6 Presentation

L5 Session

L4 Transport

L3 Network

L2 Data Link

L1 Physical

802.15.1 Bluetooth

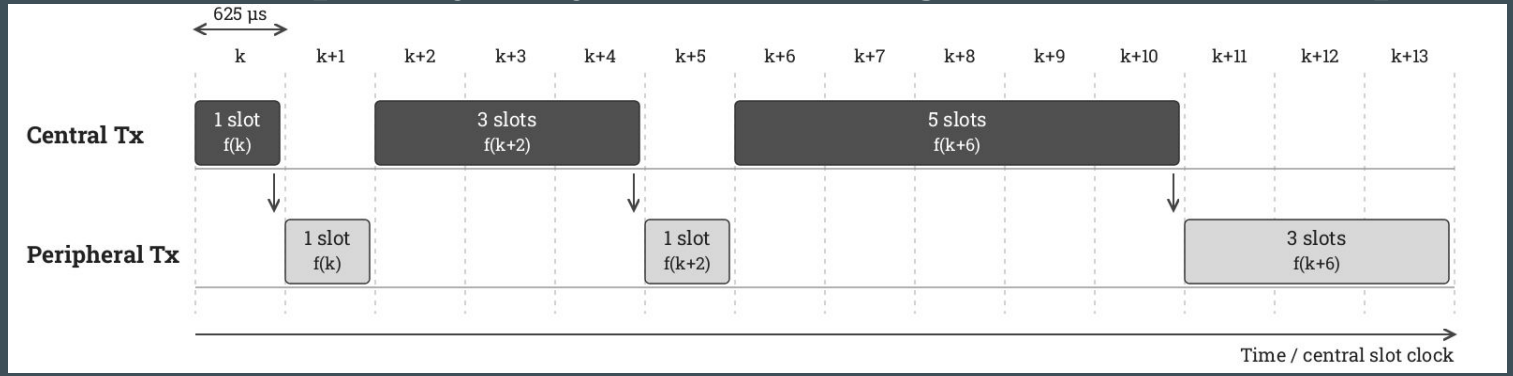
- Short range, mainly in ISM bands at 2.4 GHz
  - Same as Wi-Fi IEEE 802.11b/g
  - Up to 8 devices in a piconet (1 master and 7 slaves), up to 7 connections
  - Piconets can combine and form so-called scatternets
  - Co-located piconets are linked by sharing one master or one slave device
- Frequency Hopping Spread Spectrum (FHSS)
  - Bluetooth devices hop from one channel to another, pseudo-random, up to 1600 hops per second
  - 2.402 GHz ... 2.480 GHz divided into 79 channels with 1 MHz each

Bluetooth Classic: Timing and Scheduling

- Slotted TDD: each slot is 625 µs
  - Central starts in even slots; peripheral replies in odd slots
- Data packets may occupy 1, 3 or 5 slots
  - Carrier stays constant throughout a multi-slot packet
- FHS = Frequency Hop Synchronization packet
  - Conveys clock and address information for synchronization
- Bluetooth specifies transmission and response rules
  - The central's choice of whom and when to serve is implementation-specific

Bluetooth Classic: MAC Timing

- Bluetooth Classic: slotted time-division duplex
  - Each peripheral reply uses the preceding central channel
  - Carrier frequency stays fixed throughout a multi-slot packet.

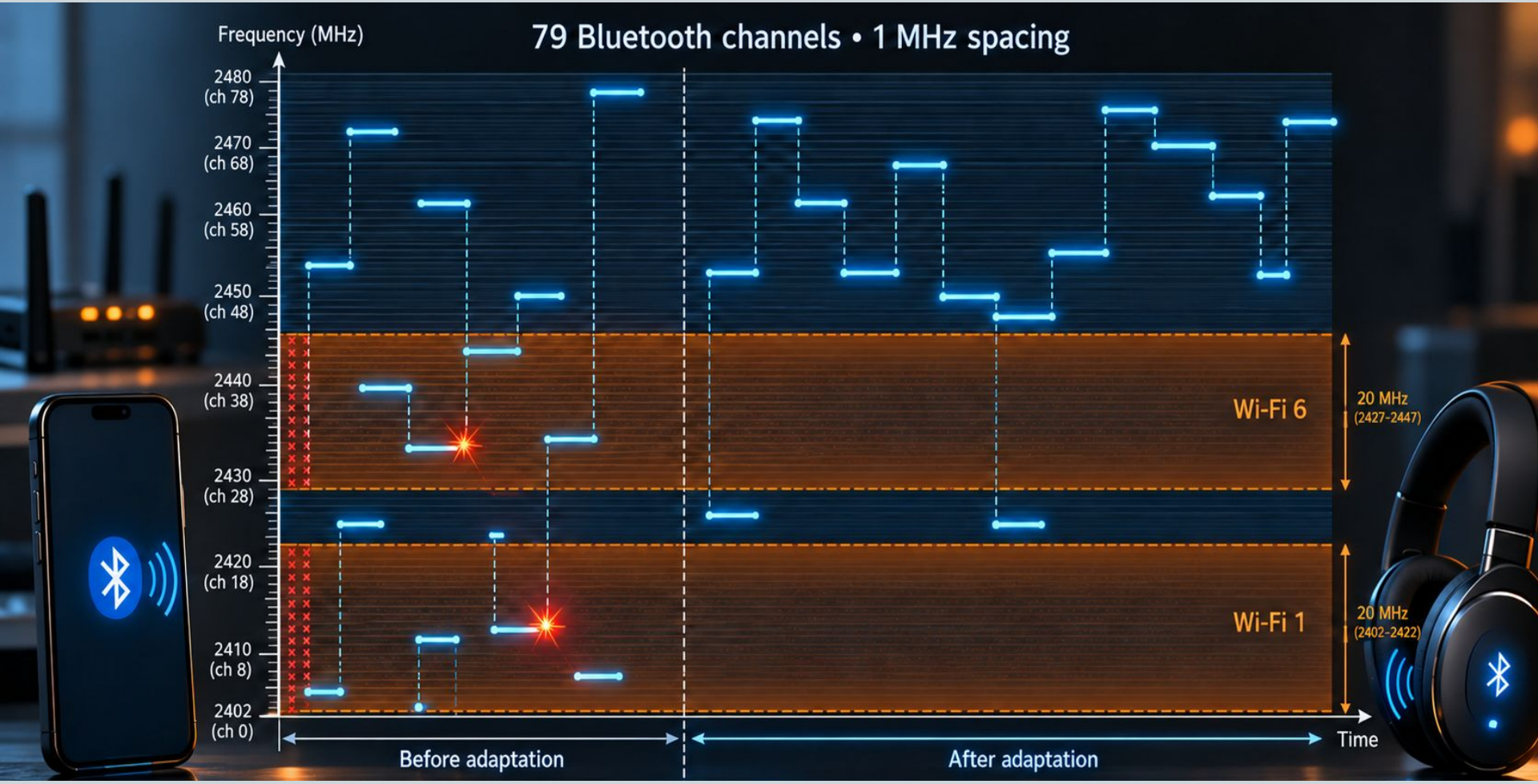


- A 3-slot or 5-slot packet occupies consecutive slots;
  - Central packets start in even slots; peripheral replies start in odd slots.
  - Packets can end before the slot boundary.

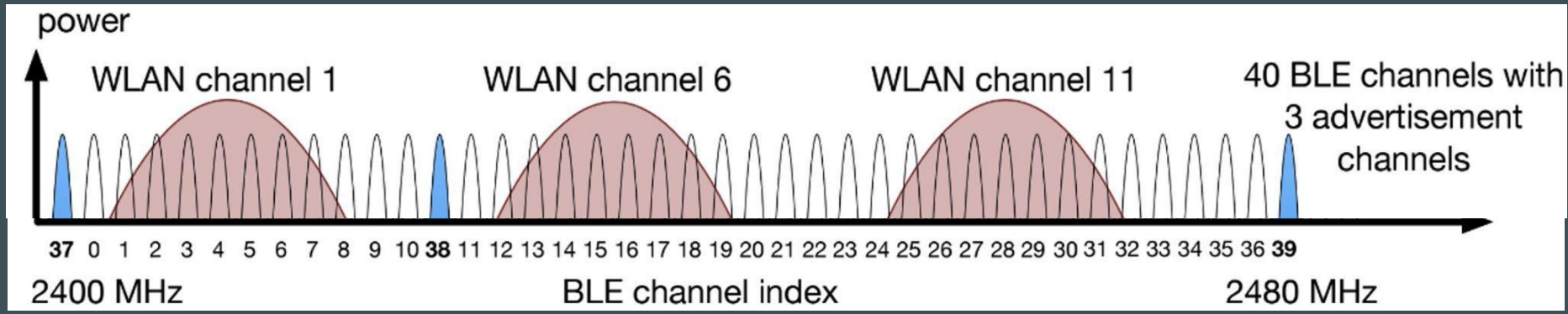
Bluetooth Versions

- Bluetooth Low Energy (BLE) was introduced in Bluetooth 4.0
- Bluetooth 5.0 added optional BLE features:
  - LE 2M for higher speed; LE Coded for longer range
  - Extended advertising: up to 255 octets per link-layer packet
- BLE: 40 channels, 2 MHz spacing, GFSK modulation
- Compatibility requires a shared transport (BR/EDR or LE) and features

Bluetooth Classic: Adaptive Frequency Hopping



BLE Radio Channels



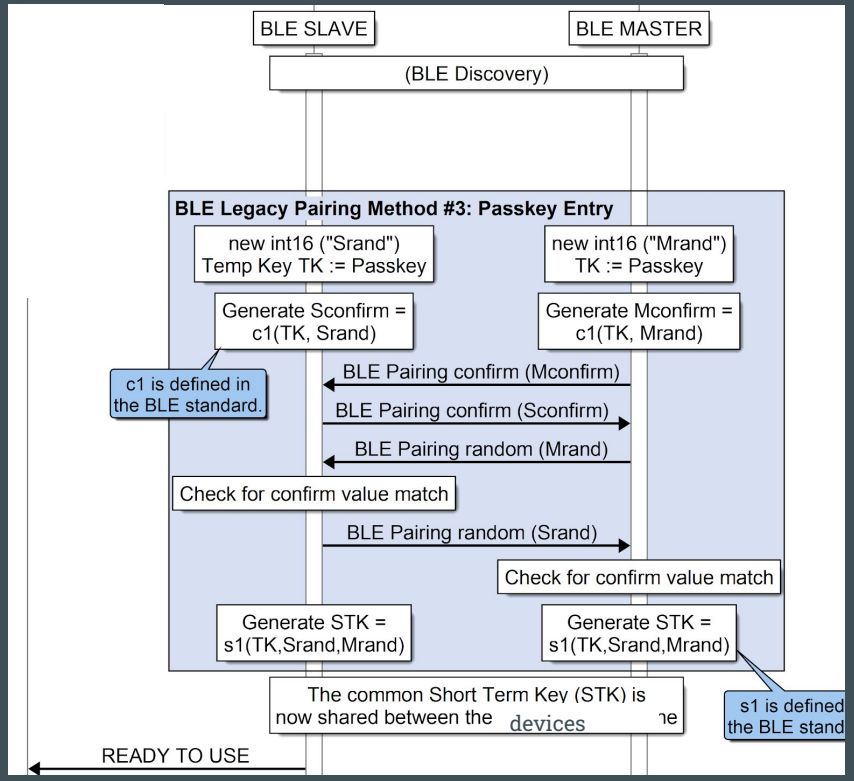
Bluetooth Low Energy (BLE)

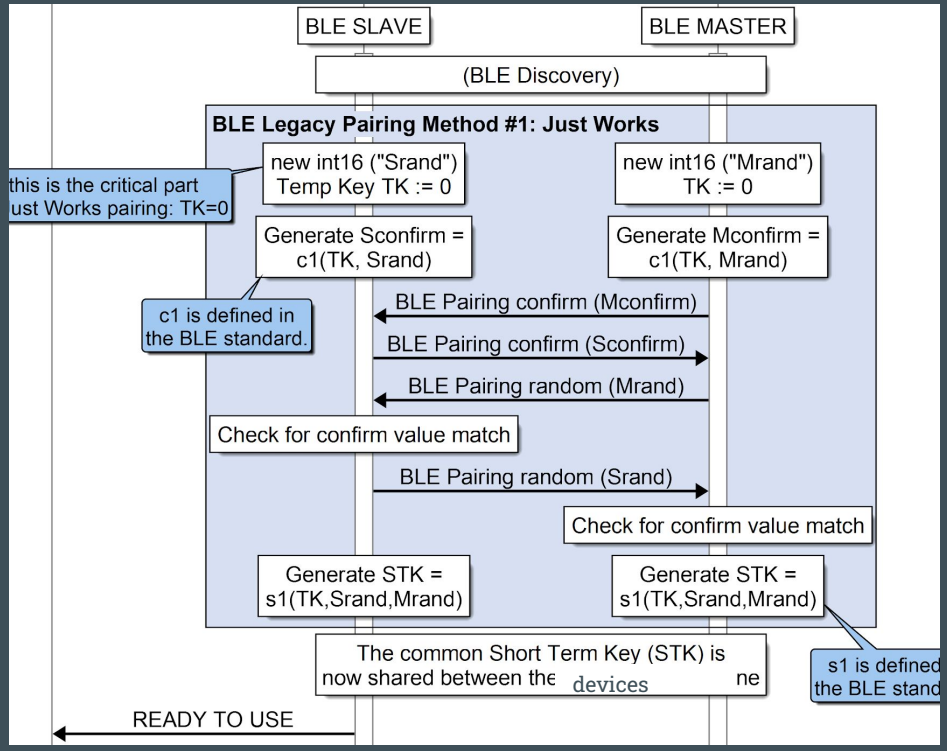
	BLE	Classic BT
Frequency Band	2.4GHz ISM Band	2.4GHz ISM Band
Frequency Channels	3 advertising and 37 data channels, 2 MHz spacing	79 channels, 1 MHz spacing
Freq. Channel Usage	Static on the three advertisement channels, optional adaptive FHSS for data exchange.	Static during advertisement, adaptive FHSS during data exchange
Modulation	GFSK	GFSK, π/4 DQPSK, 8DPSK
Energy Consumption	~0.01x to 0.5x of reference	1 (reference value)
Max Data Rate	125 Kb/s … 2 Mb/s	1 Mb/s … 3 Mb/s
Max Tx Power	1mW (0 dBm) … 100 mW (+20 dBm)	1mW (0 dBm) … 100 mW (+20 dBm)
Network Topologies	Point-to-Point (piconet), Broadcast, Mesh	Point-to-Point (piconet)

BLE Pairing: Legacy and Secure Connections

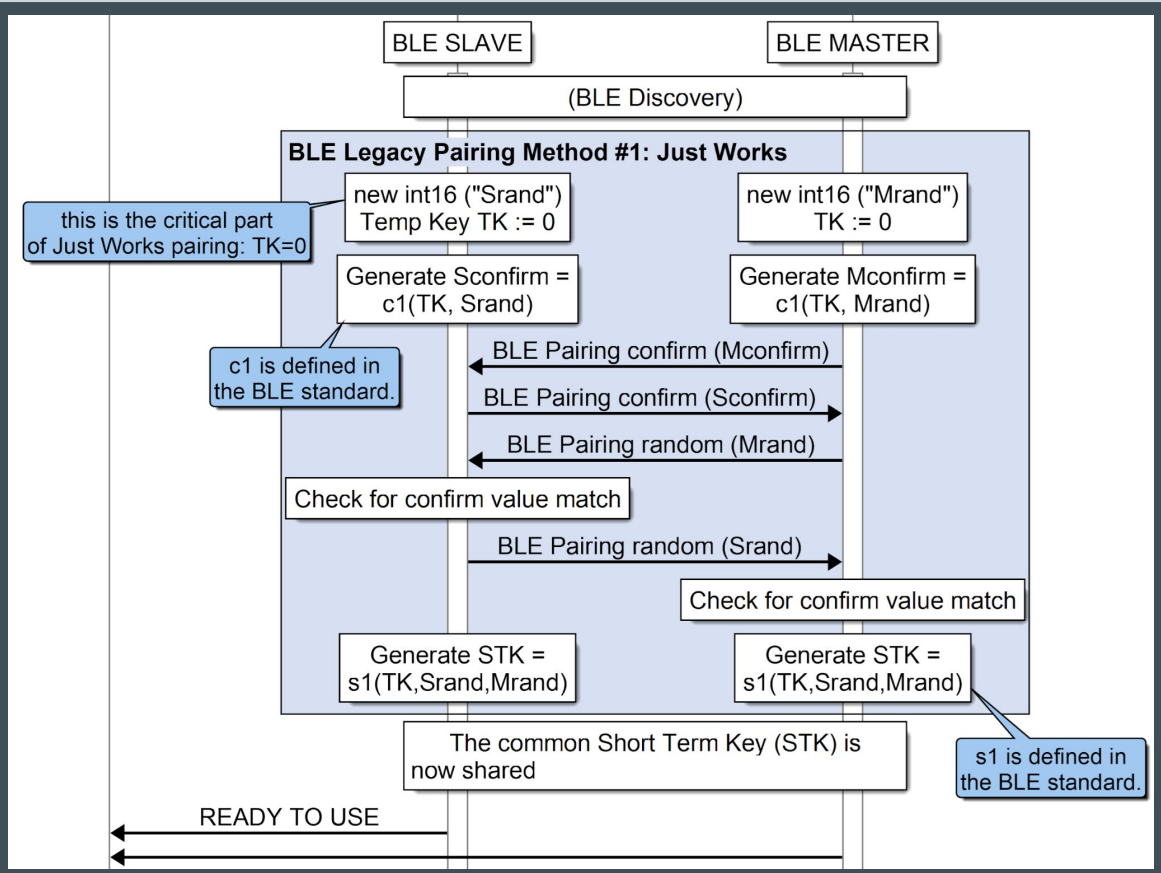
- Pairing establishes keys; bonding stores them
- LE Legacy Pairing uses a Temporary Key (TK)
  - Just Works: TK = 0; no authenticated MITM protection
- LE Secure Connections uses ECDH key agreement
  - Just Works still has no authenticated MITM protection
- Passkey or Numeric Comparison can authenticate pairing
  - Numeric Comparison requires LE Secure Connections

LE Legacy Pairing: Passkey and Just Works



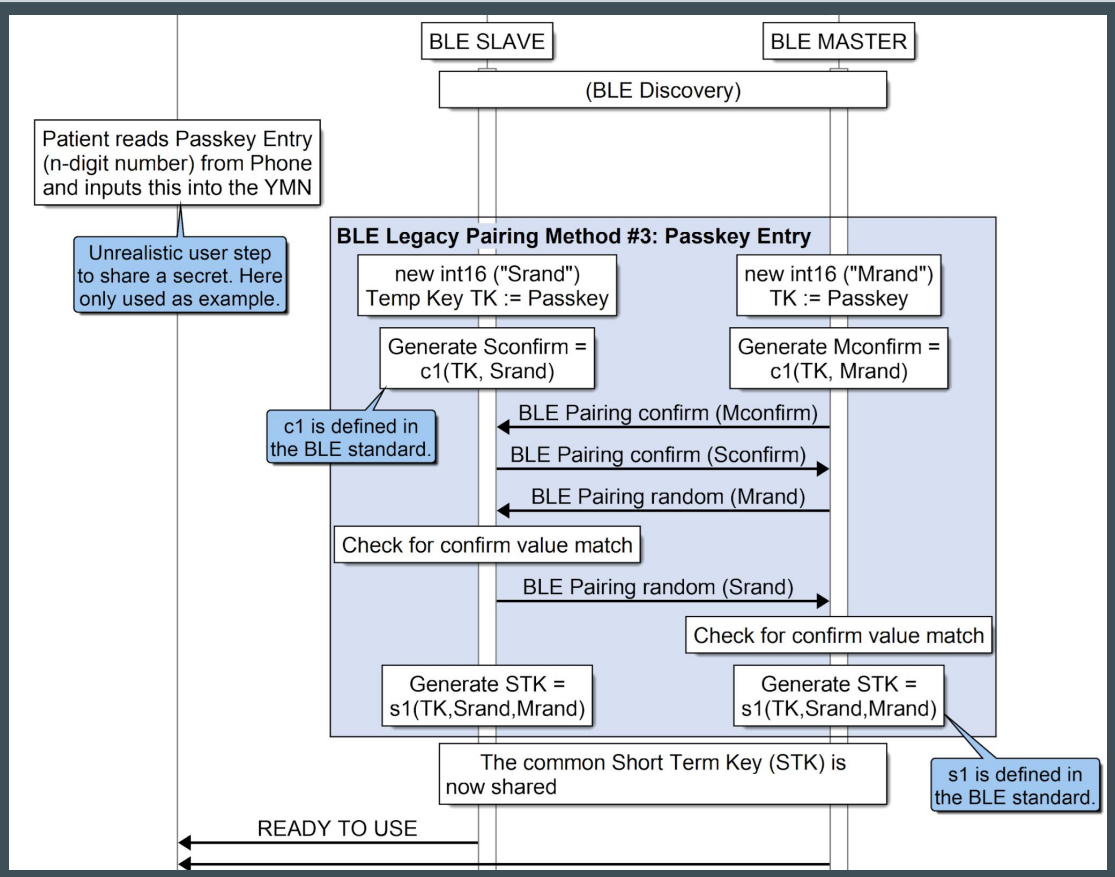


LE Legacy Pairing: Just Works



Historical TK / STK flow - not LE Secure Connections

LE Legacy Pairing: Passkey Entry



Historical TK / STK flow - not LE Secure Connections

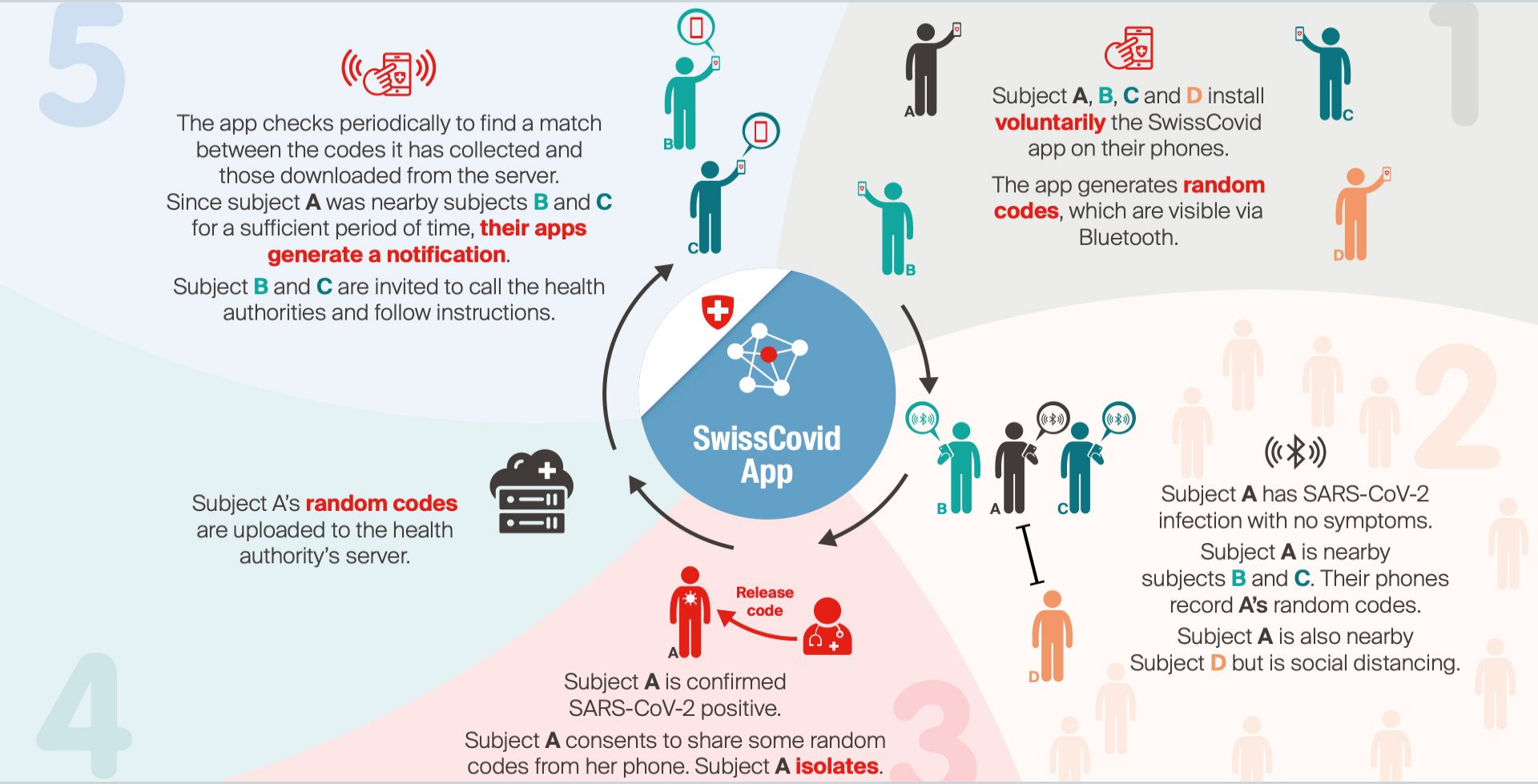
BLE Application: Location Beacons



BLE Application: Device-to-Device (Mesh) Networks



BLE Application: Covid-19 Exposure Notification



Covid-19 Exposure Notifications

- BLE advertisements are non-connectable and undirected
- Addresses and proximity identifiers rotate (~10–20 min)
  to reduce tracking; anonymity is not guaranteed
- Recommended advertising interval: 200–270 ms
- Proximity estimation uses BLE signal attenuation and
  encounter duration, without GPS location

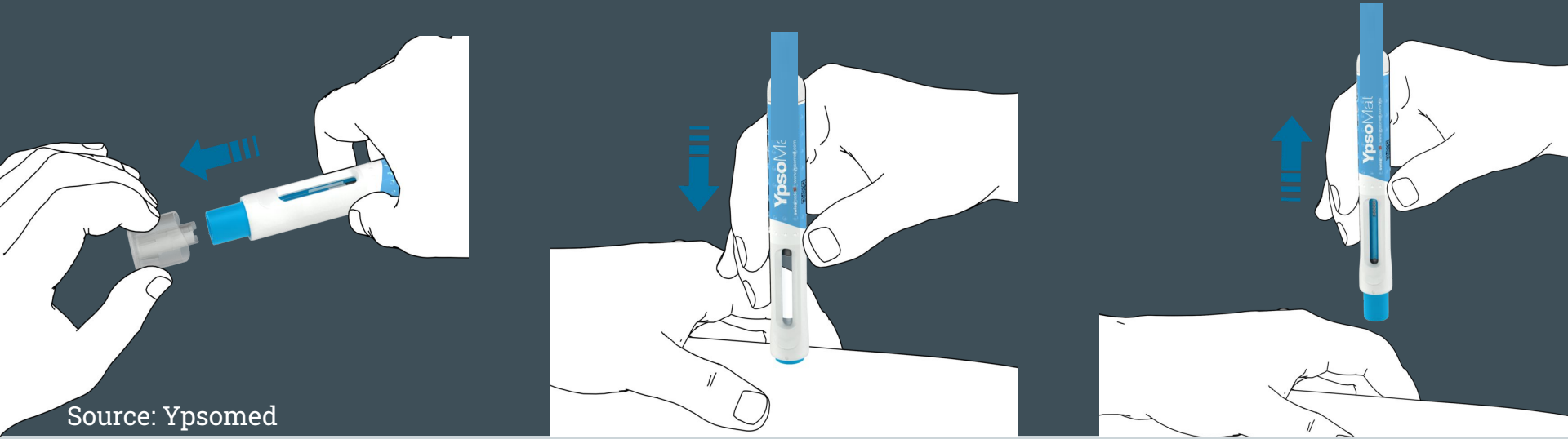
Use of BLE and VLC in a MedTech Application

- BLE for data and ranging
- VLC for activation



Self-Medication with Low-Complex Autoinjector

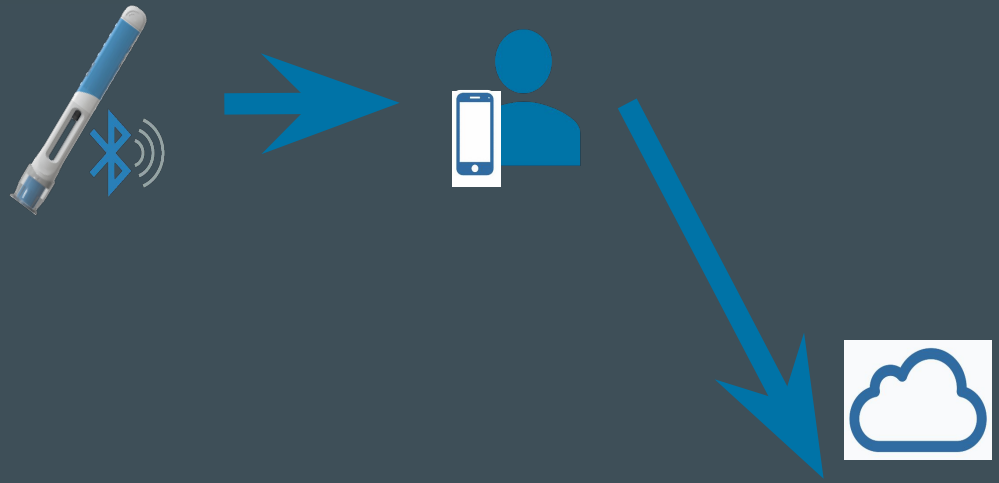
- Prefilled injection device
- Triggered by push-on-skin activation
- Acoustic confirmation clicks



Source: Ypsomed

Therapy Management

- Adherence management of self-medication
- Automated injection logging
- Assist patients, caregivers and healthcare professionals



Proximity instead of Pairing

- Broadcast injection reports to nearby phones
- RSSI indicates rough proximity; the environment affects accuracy
- Apps relay reports to the cloud
- Cloud removes duplicate reports
- Reports require authentication and verified device-patient association



Bluetooth: From Proximity to Precise Ranging

- Channel Sounding introduced in Core 6.0; version 6.3 refines ranging accuracy and efficiency
- Item finding, digital keys, proximity services
- Both devices must support the feature

Current: RSSI

Signal strength → rough proximity

Affected by obstacles and orientation

New: Channel Sounding

Phase + round-trip timing → distance

More accurate, with security measures

Bluetooth LE High Data Throughput (HDT)

- More bits per symbol: QPSK/QAM
- Error correction + rate adaptation for robustness
- Larger packets and selective block retransmission reduce overhead; application throughput is lower
- Draft October 2026

Current: BLE 2Mb/s

2 Msymbol/s × 1 bit/symbol

GFSK → 2 Mb/s PHY rate

New: HDT

2 Msymbol/s + higher-order modulation

PHY rates up to 7.5 Mb/s

CSA: Connectivity Standards Alliance

- Industry alliance: standards and product certification
  - Formerly the Zigbee Alliance; renamed in 2021
- Zigbee: mesh and application layers on IEEE 802.15.4
  - Matter: IP-based application interoperability
- Runs over Thread, Wi-Fi or Ethernet; BLE helps setup
  - A bridge can connect Zigbee devices to Matter

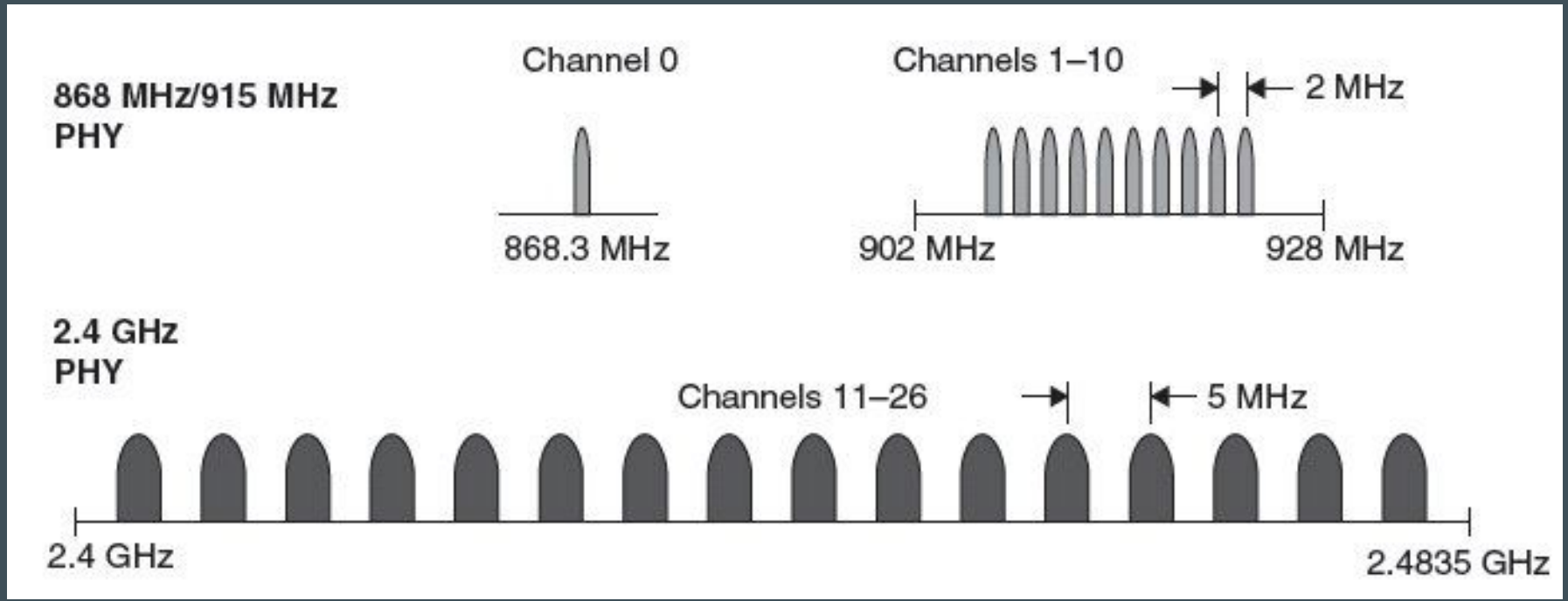
Zigbee: Low-Power Mesh on IEEE 802.15.4

- IEEE 802.15.4 defines the radio and MAC
  - Zigbee adds networking and application interoperability
- Designed for small, infrequent messages
  - Sensors, lighting and building automation
- Sleepy end devices save energy; routers relay traffic
- 2.4 GHz PHY: 250 kb/s; application throughput is lower



IEEE 802.15.4: Channel Plan (Legacy PHYs)

- 2.4 GHz: 16 channels (11–26), 5 MHz center spacing
- Sub-GHz channel plans depend on PHY and region



Zigbee: Three Device Roles

- Coordinator - forms the network
  - One per network; selects initial network parameters
- Router - forwards packets across the mesh
  - Normally stays awake; can parent end devices
- End device - communicates through one parent
  - Does not route; may sleep while its parent buffers data

IEEE 802.15.4 PHY: Selected Legacy Examples

- 2.4 GHz: DSSS + O-QPSK at 250 kb/s

PHY [MHz]	Freq. [MHz]	Chip rate [kchip/s]	Modulation	Bit rate [kb/s]	Symbol rate [ksymbol/s]
868/915	868–868.6	300	BPSK	20	20
868/915	902–928	600	BPSK	40	40
2450	2400 – 2483.5	2000	O-QPSK	250	62.5

DSSS maps data symbols to longer chip sequences.

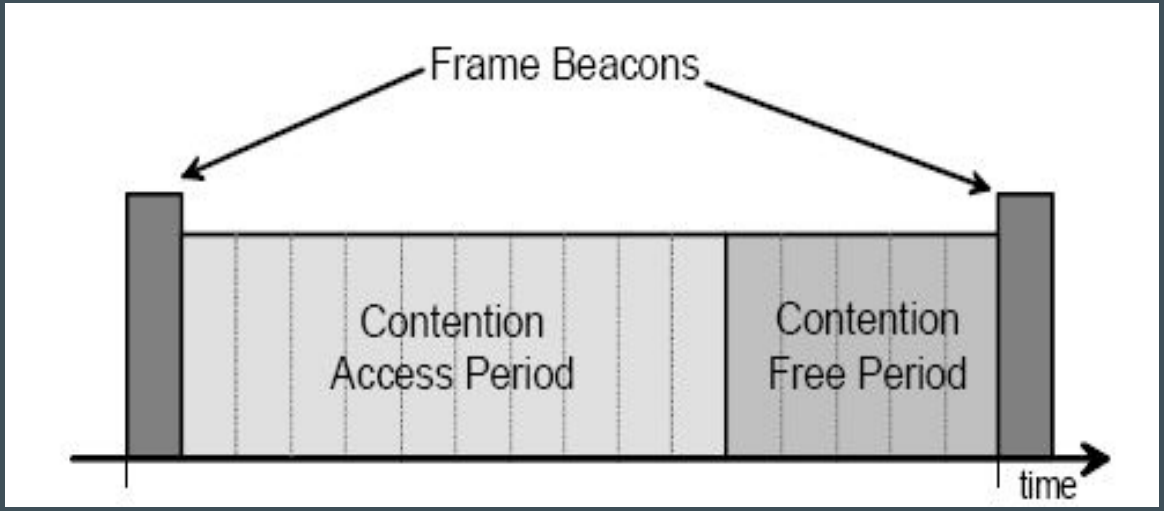
Transmit power depends on radio design and regional limits.

IEEE 802.15.4 MAC: Two Legacy Access Modes

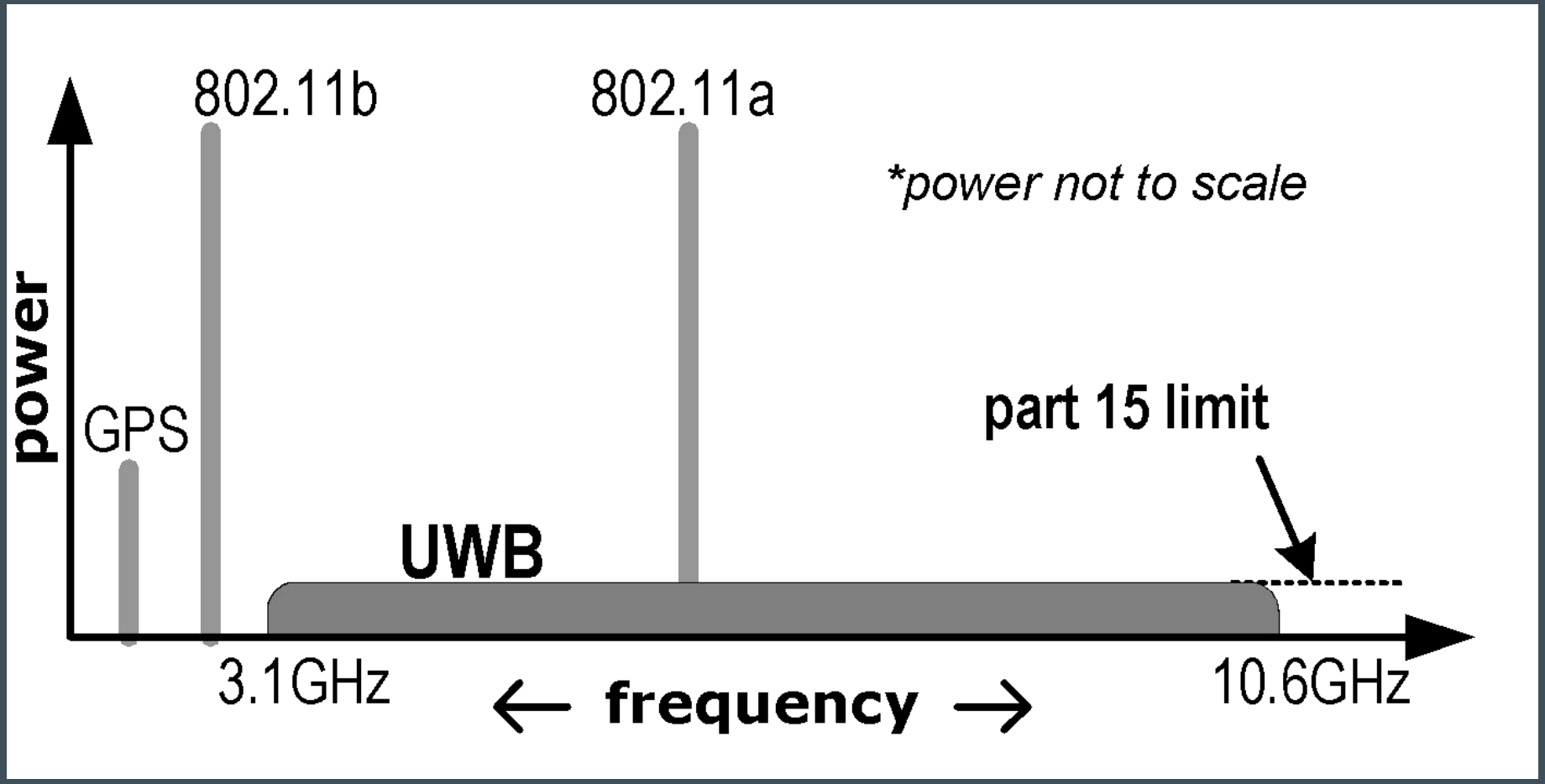
- Non-beacon-enabled: unslotted CSMA-CA
  - Typical conventional Zigbee operation; no periodic superframe
- Beacon-enabled: coordinator defines a superframe
  - CAP: slotted CSMA-CA; optional CFP: reserved GTS
- Acknowledgements and retries support reliable delivery
- Sleep saves energy; polling and buffering add delay

IEEE 802.15.4: Beacon-Enabled Superframe

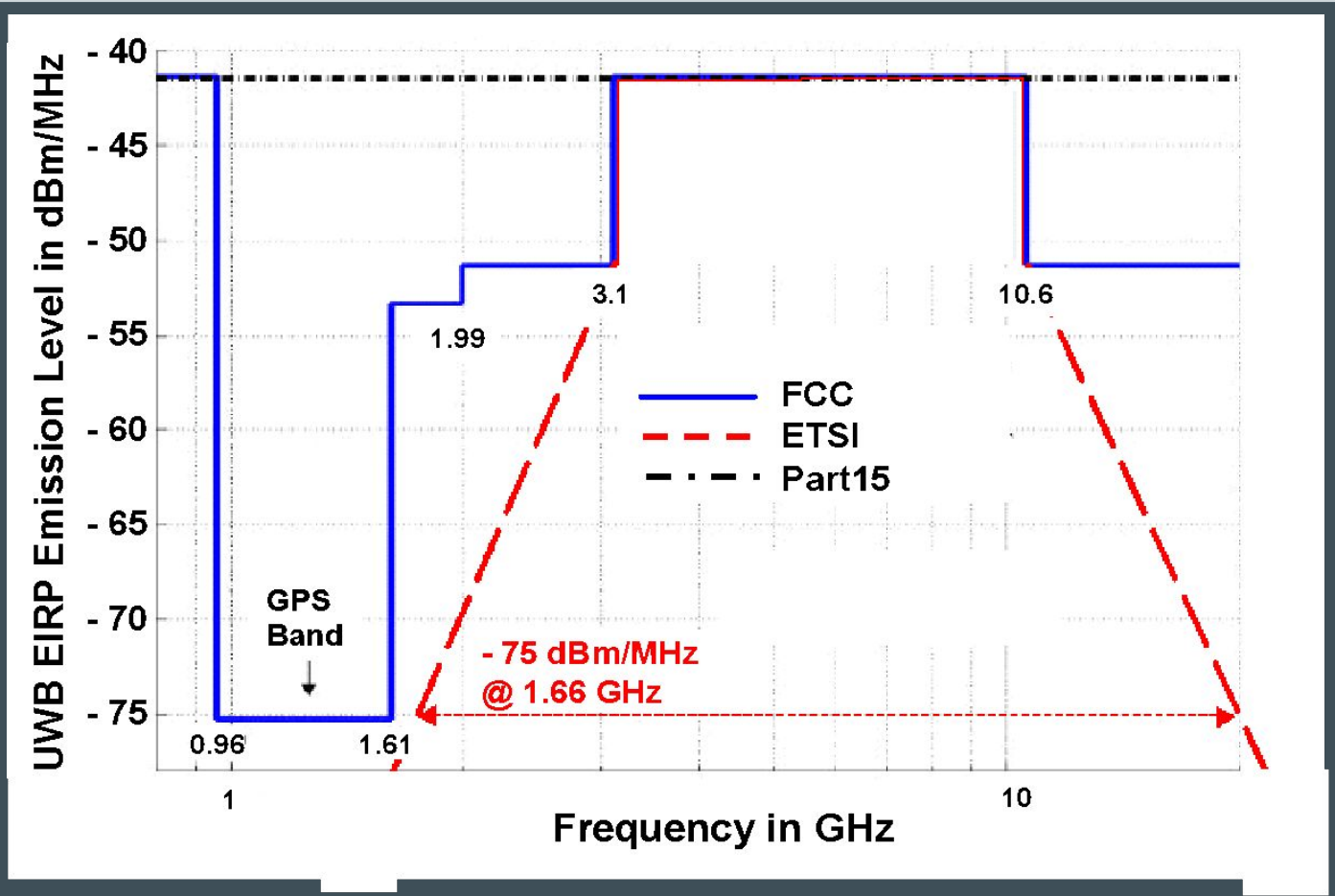
- CAP: compete using slotted CSMA-CA
- CFP: optional reserved slots (GTS); an inactive period may follow



Ultra Wideband Communications, UWB (“underlay”)



UWB Regulation



802.15.3a: UWB Proposals (Withdrawn)

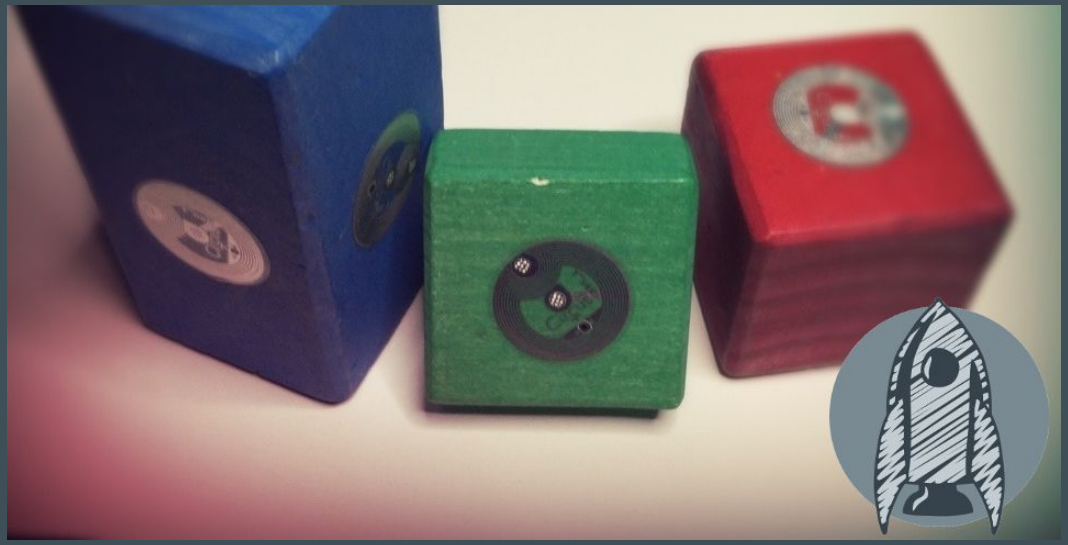
	DS-UWB	MB-OFDM
PHY technique	Single-carrier direct sequence spreading	Multiband-OFDM
Frequency management	Two bands (lower band: 3.1 ~ 4.85 GHz Higher band: 6.2 ~ 9.7 GHz)	Four frequency band groups. Each band covers 528 MHz.
Data rates	28, 55, 110, 220, 500, 660, 1000 and 1320 Mb/s	53.3, 80, 106.7, 160, 200, 320, 400 and 480 Mb/s
Typical modulation	BPSK	QPSK
Channel coding	Convolutional coding with interleaving	Convolutional coding with interleaving
Logical channels	Total 12 channels	Total 18 channels through Multiplexing

UWB in Practice: NXP Chip Demo



NFC: Tap to Connect

- Tap to exchange data over a few centimeters
- Passive tags harvest power from the reader
- Payments, access control and product information



NFC: How It Works

- 13.56 MHz: magnetic near-field coupling
- Reader powers a passive tag; tag modulates its load
- Modes include reader/writer and card emulation
- Data rate and memory depend on technology and tag
- Short range alone does not guarantee security
- NFC Forum + ISO/IEC 14443, 18092 and 15693

Internet of Things

- Wi-Fi HaLow, BLE, ZigBee
  - Wi-Fi Alliance
  - Bluetooth SIG
  - Connectivity Standards Alliance, https://csa-iot.org
  - IEEE 802.15.4g - Utility (Smart Meter) Industry
- LoRaWAN: open standard (LoRa Alliance)
  - LoRa radio: proprietary technology (Semtech)
- LTE-M, Narrow Band IoT (NB-IoT), Cellular 5G
  - 3GPP, cellular infrastructure

